

Endocrine disrupting nonylphenols are ubiquitous in food.

4-Nonylphenols (NPs) are common products of biodegradation of a widely used group of nonionic surfactants, the nonylphenol ethoxylates (NPEs). These compounds are known to be persistent, toxic, and estrogen active. There is a worldwide scientific and public discussion on the potential consequences of human long term dietary exposure to such endocrine disrupters. Despite numerous determinations of NPs in environmental samples no systematical reports exist relating to concentrations of NPs in food. We analyzed NPs in 60 different foodstuff commercially available in Germany. The results indicate that NPs are ubiquitous in food. The concentrations of NPs on a fresh weight basis varied between 0.1 and 19.4 microg/kg regardless of the fat content of the foodstuff. Based on data on German food consumption rates and these first analyses of NPs in food, the daily intake for an adult was calculated to be 7.5 microg/day NPs. For infants exclusively fed with breast milk or infant formulas daily intakes of 0.2 microg/day and 1.4 microg/day NPs, respectively, can be estimated.